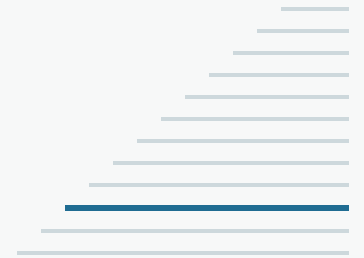

RepoRevive: Legacy Code, Revived.

AI-guided health audit and renovation for legacy repositories — a multi-agent pipeline that scores, plans, and fixes real code.

AUDIT → PLAN → RENOVATE | HEALTH SCORE 0-100 | 4 AGENTS • ONE TEAM



Legacy code is where velocity goes to die.

Most of the world's software is legacy. Nobody knows which files are safe to touch — so teams freeze, rewrite, or gamble.

WEEKS

To onboard

onto an unfamiliar repository

01

0%

Test coverage

in much of the code you inherit

02

HIDDEN

CVEs

inside outdated dependency
trees

03

NOISE

Static tools

dump warnings — not fixes

04

Three orchestrated phases. One live health score.

01



AUDIT

The Inspector agent scans the full repository and produces a 0–100 health score, an architecture diagram, and a risk heatmap.

PHASE 01

02



PLAN

The Architect agent converts findings into a prioritized, dependency-ordered backlog where every task carries acceptance criteria.

PHASE 02

03



RENOVATE

The Executor completes tasks one by one, the Validator reviews every change, and the score climbs live.

PHASE 03

 Paste a [GitHub URL](#) — watch the score climb.

PHASE 1 • AUDIT

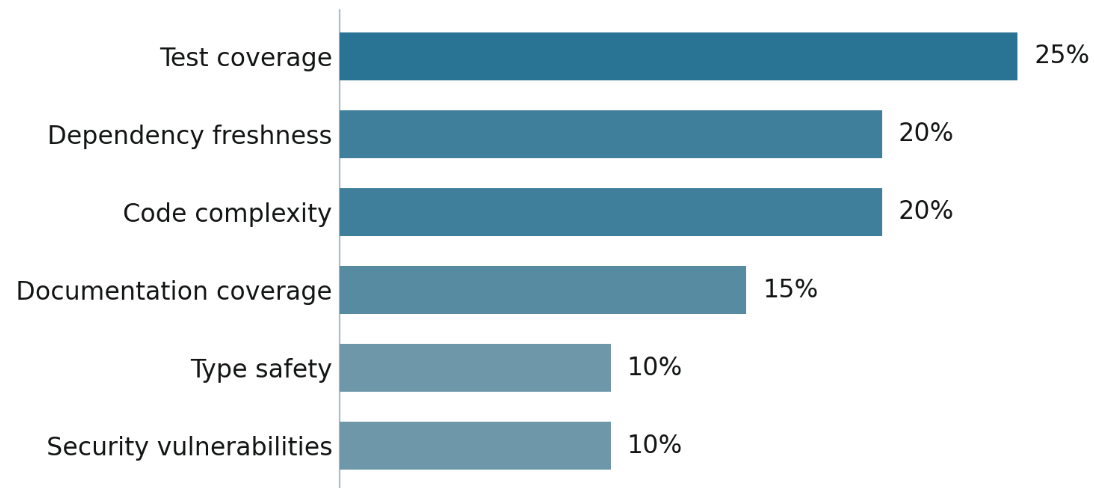
The Inspector agent scores everything.

Tree-sitter AST parsing, dependency scanning, coverage and complexity metrics, documentation checks — merged by Bob 2.0 into structured JSON findings that feed a transparent six-dimension scoring engine.

■ 0-100 SCORE

■ RISK HEATMAP

■ ARCH DIAGRAM



Six weighted dimensions drive the health score.

The Architect agent turns findings into a build order.

Every task is small, verifiable, and dependency-ordered — with rationale, affected files, and acceptance criteria. Approve, reorder, or descope from the dashboard. Renovation never becomes a rewrite.

T-014

Add integration tests for auth middleware

+6 PTS

2 FILES

AC: COVERAGE ≥ 80%

T-015

Upgrade express 4.17 → 4.21

+4 PTS

1 FILE

AC: ROUTES PASS

T-016

Extract config loader from monolith

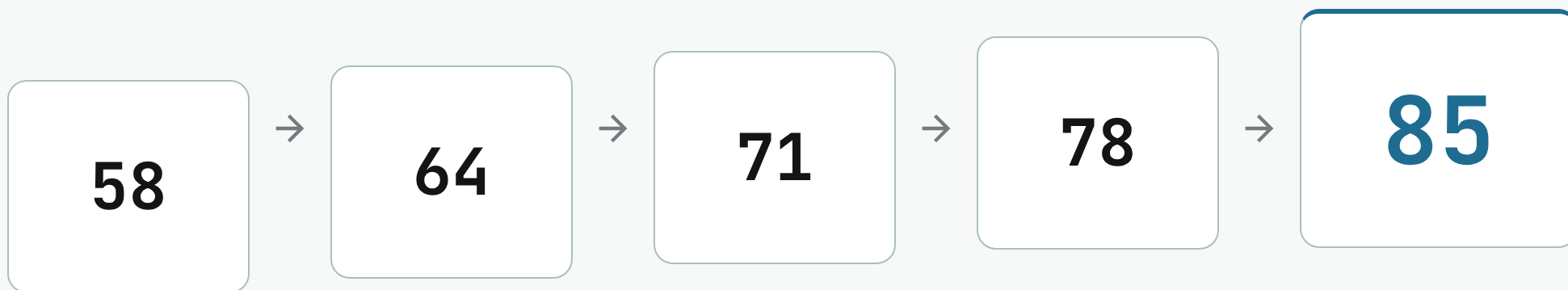
+3 PTS

4 FILES

AC: IMPORTS CLEAN

Watch the score climb with every verified fix.

The Executor lands one task at a time, running builds and tests through its shell. The Validator reviews each diff before commit, and the health score recomputes after every merge — progress you can measure, not guess.



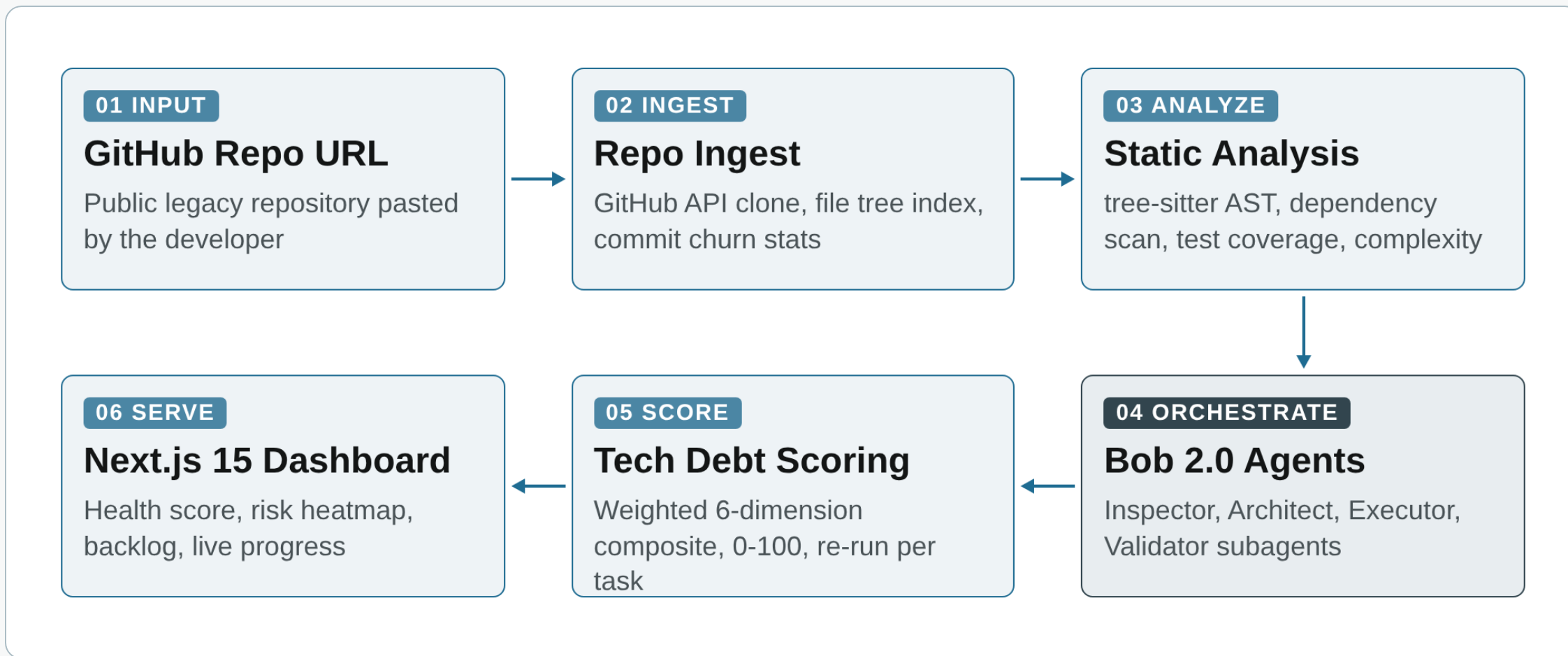
Health score across five merged tasks

EXECUTOR

VALIDATOR

LIVE ACTIVITY FEED

From GitHub URL to live dashboard — Bob at the hub.



Ingest → analyze → orchestrate → score → serve. IBM Bob 2.0 coordinates Inspector, Architect, Executor and Validator.

Bob 2.0 is the orchestration core — not a chat sidebar.



Agent Mode + Full-Repo Context

The Inspector reasons over the whole codebase, not isolated snippets.



Parallel Subagents

Dependency, complexity and documentation audits run simultaneously.



Document Understanding

Reads READMEs, lockfiles and CI configs as evidence.



Shell & Tool Use

The Executor runs builds and tests; the Validator reviews diffs.

Maintenance is a cost center. Make it a pipeline.

WHO IT SERVES

 Enterprise maintenance teams buried in legacy services

 Agencies inheriting client codebases

 Open-source maintainers drowning in issues

 New hires onboarding onto unfamiliar repos

WHAT IT RETURNS

 Minutes of automated audit replacing weeks of archaeology

 Score points per task — ROI attached to every fix

 Verified, test-backed changes instead of vibes

 A health baseline you can monitor continuously

What you are about to see.

1

Paste a real GitHub URL — the Inspector audits it live; the activity feed and risk heatmap render as agents work.

2

The Architect proposes the plan — we approve a task with one click.

3

The Executor lands the fix — the health score climbs on screen, in real time.

Every recommendation is traceable to specific code evidence.

EXECUTION

48 hours, four builders, one loop.

FRI 15:00

Scaffold repo, GitHub ingestion, Bob agent harness

FRI 20:00

Scoring engine + risk heatmap shipped

SAT AM

Inspector and Architect wired into the dashboard

SAT 20:00

Executor + Validator renovation loop closes

SUN 06:00

End-to-end demo passes on real repositories

SUN 12:00

Video, deck and submission — 3-hour buffer

RepoRevive. **Legacy code, revived.**

Built with IBM Bob 2.0 at the lablab.ai Bob 2.0 Hackathon — September 2026.

[PUBLIC REPO — SEE PER-MEMBER BOB SESSION SCREENSHOTS](#)

[DEMO — LINK IN SUBMISSION](#)

Team RepoRevive — four builders, forty-eight hours.